

Results :



Frequency Response of 2 Stage OTA (Differential signals)

DC Gain : 80db

Unity Gain BW ~ 5.18M Hz

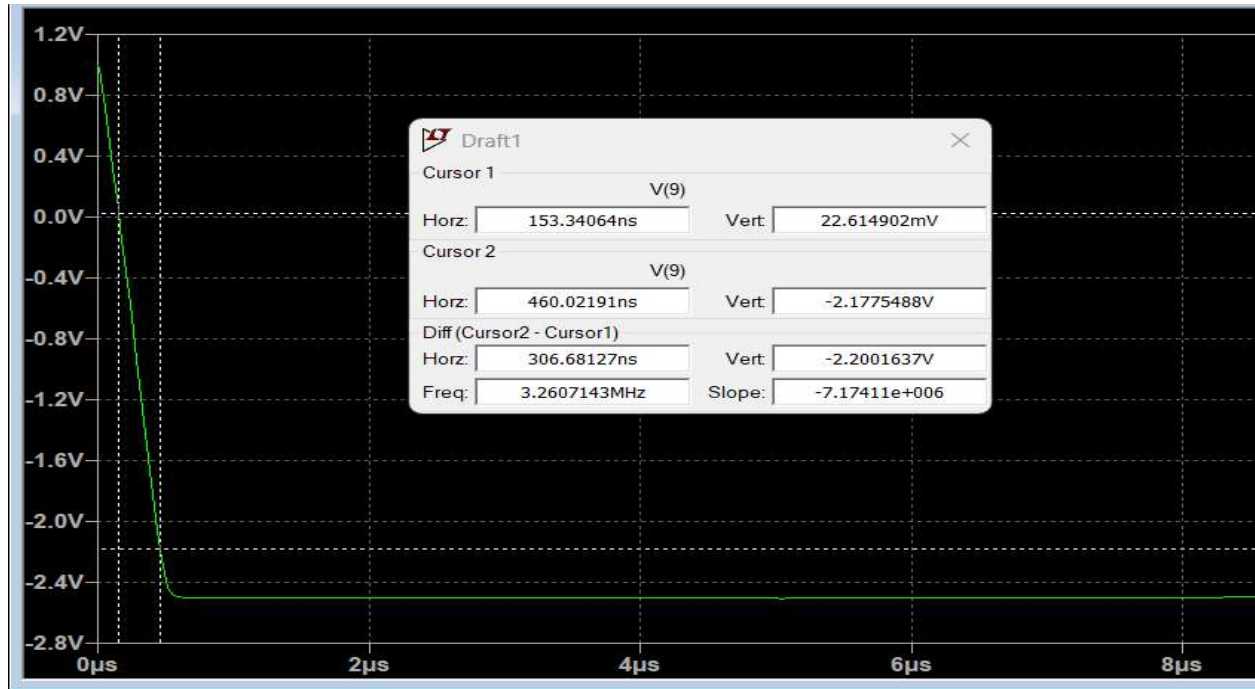
Phase Margin (PM) : $180 - \text{Phase(UGB)} \Rightarrow 180 - |-113| = 67 \text{ Degrees}$



Frequency Response of 2 Stage OTA (Common Mode Signals)

Common mode Gain is about -7.15db

Common mode rejection Ratio (CMRR) is $80 - (-7.15) = 87.15\text{db}$



Cursor 1: $t = 153.34\text{ns}$, $V = 22.61\text{mV}$ ($\approx$ top of fall)

Cursor 2: $t = 460.02\text{ns}$, $V = -2.1775\text{V}$ ($\approx$ bottom of fall)

$\Delta t = 306.68\text{ns}$

$\Delta V = -2.2001\text{V}$

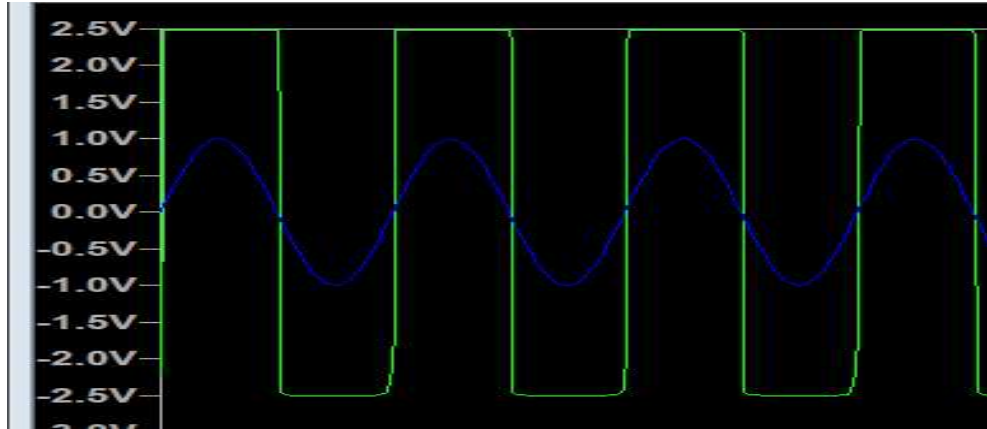
Slope = $-7.17411 \times 10^6 \text{ V/s}$

SR = $7.17 \text{ V}/\mu\text{s}$ (falling)

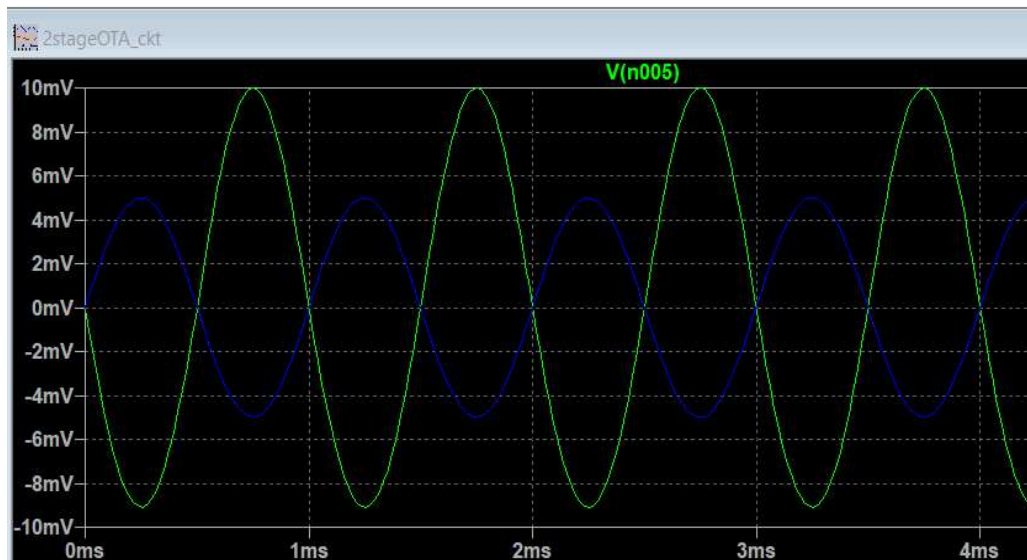
SR_theoretical = $I_{\text{ref}} / C_c = 30\mu\text{A} / 3\text{pF} = 10 \text{ V}/\mu\text{s}$

SR_simulated = $7.17 \text{ V}/\mu\text{s}$

- pSome simple circuits to check the working of OTA



OTA as Comparator



OTA as Inverting amplifier (Gain = 2)